

# Skills Summary

## Ratios with Tape Diagrams and Double Number Lines

*Use this reference while completing your worksheet or when studying.*

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### Sharing a Total with a Tape Diagram

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**Rule:** Draw one row of boxes per person, matching the ratio. Then: add up **all** the boxes, divide the total by that count to find the value of **one box**, and multiply by each person's boxes.

**Example:** Two friends share 30 dollars in the ratio 2:3. How much is the smaller share?

**Step 1.** The tape has  $2 + 3 = 5$  equal boxes for 30 dollars, so one box is worth  $30 \div 5 = 6$  dollars.

**Step 2.** The smaller share has 2 boxes:  $2 \times 6 = 12$  dollars, so it is **12** dollars.

**Try it:** Two teams share 28 stickers in the ratio 3:4. How many stickers does the **larger** team get?

check: 19

### Differences with a Tape Diagram

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**Rule:** When a problem gives the **difference** between two quantities, count the **extra** boxes in the longer tape. Divide the difference by the extra boxes to find the value of one box.

**Example:** Two jars hold marbles in the ratio 5:3, and the first jar holds 10 more. How many marbles are in the first jar?

**Step 1.** The first tape has  $5 - 3 = 2$  extra boxes, and those hold the 10 extra marbles, so one box is  $10 \div 2 = 5$  marbles.

**Step 2.** The first jar has 5 boxes:  $5 \times 5 = 25$ , so it holds **25** marbles.

**Try it:** Two shelves hold books in the ratio 7:4, and the first shelf holds 9 more. How many books are on the first shelf?

check: 21

**(!) Watch out:** Divide the difference by the **extra** boxes only — never by the total number of boxes, and never by the number of people.

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### Reading a Double Number Line

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**Rule:** On a double number line, marks that line up show equal ratios. To reach a new mark, multiply or divide **both** rows by the same number — a handy first move is dividing down to the mark for **one** unit.

**Example:** A walker covers 8 km every 2 hours. How far in 5 hours?

**Step 1.** Divide both rows by 2 to get the one-hour mark:  $8 \div 2 = 4$  km per hour.

**Step 2.** Multiply the one-hour mark by 5:  $4 \times 5 = 20$ , so she walks **20** km.

**Try it:** A printer prints 12 pages every 3 minutes. How many pages in 7 minutes?

check: 87

### Key Vocabulary

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**Tape diagram:** rows of equal boxes showing a ratio      **Unit value:** what one box (or one unit mark) is worth

**Double number line:** two aligned scales growing in the same ratio      **Difference:** the amount by which one quantity exceeds another